

Topology Final Exam Spring 2020

Please choose four problem and solve them. I will be grading the four problem you choose.

1. (20) Classify all connected 2-sheeted and 3-sheeted covers of $S^1 \vee S^1$, up to isomorphism of covering spaces without base points. Compute the deck transformation groups in each case and indicate which covers are normal.
2. (20) Determine which abelian groups A fit into the exact sequence $0 \rightarrow \mathbb{Z}_{p^n} \rightarrow A \rightarrow \mathbb{Z}_{p^n} \rightarrow 0$. Answer the same question for $0 \rightarrow \mathbb{Z} \rightarrow A \rightarrow \mathbb{Z}_{p^n} \rightarrow 0$.
3. (20) Show that $S^1 \times S^1$ and $S^1 \vee S^1 \vee S^2$ have isomorphic homology groups in all dimension but their universal covering spaces do not.
4. (20) Show that the quotient map $S^1 \times S^1 \rightarrow S^2$ collapsing the subspace $S^1 \vee S^1$ to a point is not nullhomotopic (use homology). On the other hand, show that any map $S^2 \rightarrow S^1 \times S^1$ is nullhomotopic.
5. (20) Let us define space $X := M(\mathbb{Z}_m, n)$ by attaching a cell e^{n+1} to S^n with degree m map. First compute the homology and cohomology groups of X . Then
 - show that quotient map $X \rightarrow X/S^n = S^{n+1}$ induces the trivial map on $\tilde{H}_i(-; \mathbb{Z})$ for all i but not on $H^{n+1}(-; \mathbb{Z})$.
 - show that the inclusion $S^n \rightarrow X$ induces the trivial map on $\tilde{H}^i(-; \mathbb{Z})$ for all i but not on $H_n(-; \mathbb{Z})$.
6. (20) Let X be $\mathbb{CP}^2$ with a cell e^3 attached by a map $S^2 \rightarrow \mathbb{CP}^1 \subset \mathbb{CP}^2$ of degree p (p is prime). Let $Y = M(\mathbb{Z}_p, 2) \vee S^4$, see the previous problem for construction of $M(\mathbb{Z}_p, 2)$. Show that X and Y have isomorphic cohomology rings with $\mathbb{Z}$ coefficients but not with $\mathbb{Z}_p$ coefficients.